18 (a) Work out the value of $(2\sqrt{5})^2$

$(2\sqrt{5})^2 = 2^2 \times (\sqrt{5})^2 = 4 \times 5 = 20$

1 mark
Square both factors.

20

(1)

(b) Show that $(1 - \sqrt{7})(3 + 2\sqrt{7})$ can be written in the form $a + b\sqrt{7}$ where a and b are integers.

$(1 - \sqrt{7})(3 + 2\sqrt{7})$

$= 3 + 2\sqrt{7} - 3\sqrt{7} - 14$

$= -11 - \sqrt{7}$

$a = -11, b = -1$

2 marks total
Expand four products.
$\sqrt{7} \times \sqrt{7} = 7$.

(2)

(c) Rationalise the denominator of $\dfrac{1}{4 - \sqrt{11}}$

$\dfrac{1}{4 - \sqrt{11}} \times \dfrac{4 + \sqrt{11}}{4 + \sqrt{11}} = \dfrac{4 + \sqrt{11}}{16 - 11} = \dfrac{4 + \sqrt{11}}{5}$

2 marks total
Multiply top and bottom by the conjugate, $4 + \sqrt{11}$.

(2)

(Total for Question 18 is 5 marks)

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